

September 2, 2026

Lab Task 1: Perceptron for Binary Classification Without Using Built-in Libraries

Objective:

Implement the Perceptron learning algorithm from scratch for binary classification.

Tasks:

1. Load a binary classification dataset.
2. Perform the necessary preprocessing.
3. Initialize the weights and bias.
4. Implement the weighted sum and step activation function.
5. Implement the Perceptron weight update rule.
6. Train the Perceptron for a specified number of epochs.
7. Predict the class labels for the test samples.
8. Compare the predicted labels with the actual labels.
9. Analyze the effect of different learning rates and numbers of epochs.

Suggested Dataset:

Iris dataset by selecting any two classes.

Lab Task 2: Perceptron for Binary Classification Using a Built-in Library

Objective:

Apply a built-in Perceptron classifier for binary classification and compare its performance with the implementation developed from scratch.

Tasks:

1. Load a binary classification dataset.
2. Perform the necessary preprocessing.

3. Split the dataset into training and testing sets.
4. Train a Perceptron classifier using a machine learning library.
5. Predict the class labels for the test samples.
6. Compare the predicted labels with the actual labels.
7. Compare the results with the Perceptron implementation developed from scratch.

Suggested Library:

Scikit-learn.

Lab Task 3: Logistic Regression for Binary Classification Using a Built-in Library

Objective:

Build and apply a Logistic Regression model for binary classification.

Tasks:

1. Load a binary classification dataset.
2. Separate the input features and target labels.
3. Perform the necessary preprocessing.
4. Split the dataset into training and testing sets.
5. Train a Logistic Regression model using a machine learning library.
6. Predict class probabilities and class labels.
7. Compare the predicted labels with the actual labels.
8. Analyze the effect of different classification thresholds on the predicted labels.

Suggested Dataset:

Breast Cancer Wisconsin dataset.

Lab Task 4: Logistic Regression from Scratch

Objective:

Implement Logistic Regression from scratch using gradient descent and compare its results with a library implementation.

Tasks:

1. Load a binary classification dataset.
2. Perform the necessary preprocessing.

3. Initialize the weights and bias.
4. Implement the sigmoid activation function.
5. Compute the predicted probabilities.
6. Calculate the prediction error.
7. Update the weights and bias using gradient descent.
8. Repeat the training process for a specified number of iterations.
9. Predict class labels for the test samples.
10. Compare the results with a built-in Logistic Regression model.